Theory of Computation

Unit 1 – Lecture 2
Alphabet, Symbols, Strings & Empty String

Department of Computer Science & Engineering

B.Tech (CSE/IT)

Learning Objectives

By the end of this lecture, students will be able to:

- Understand the concept of Alphabet in Formal Language Theory.
- Define symbols and distinguish them from alphabets.
- Explain strings and their mathematical representation.
- Understand different types of strings.
- Analyse the Empty String (ϵ) and its importance.
- Perform basic string operations.

Computation
A process of transforming input into meaningful output.

Automata
An abstract mathematical machine that reads input strings and produces:

- Acceptance
- Rejection

Formal Language
A collection of strings generated using predefined rules.

$$L = \{w \mid w \text{ satisfies a given condition}\}$$

Where

- L : Language
- w : String

Alphabet (Σ)

Definition

An alphabet is a finite, non-empty set of symbols used to construct strings and formal languages.

Represented by

$$\Sigma$$

Characteristics

- Contains symbols
- Finite in size
- Symbols are distinct
- Alphabet itself is not a language

Examples

$$\Sigma = \{0, 1\}$$

$$\Sigma = \{a, b, c, \dots, z\}$$

$$\Sigma = \{A, C, G, T\}$$

Examples of Alphabets

Application	Alphabet
Binary System	$\{0, 1\}$
DNA Sequence	$\{A, C, G, T\}$
Programming Languages	Letters, Digits, Operators

Applications

- Digital Systems
- Bioinformatics
- Compiler Design
- Programming Languages

Finite and Infinite Alphabets

Finite Alphabet
Examples

$$\Sigma = \{0, 1\}$$

$$\Sigma = \{a, b, c\}$$

Infinite Alphabet
Example

$$\Sigma = \{x_1, x_2, x_3, \ldots\}$$

Important

Automata Theory mainly uses finite alphabets because computers process fixed sets of input symbols.

What is a Symbol?

Definition
A symbol is the smallest individual element of an alphabet.
Example

$$\Sigma = \{0, 1\}$$

Then

$$0,\ 1$$

are symbols.

Important Points

- Single character
- Belongs to an alphabet
- Used to construct strings

Difference Between Alphabet and Symbol

Alphabet	**Symbol**
Collection of symbols	Single element
Represented by Σ	Represented by a character
Contains many symbols	Only one symbol
Used to generate strings	Used to construct strings

Example

$$\Sigma = \{a, b, c\}$$

Alphabet: Σ
Symbols: a, b, c

What is a String?

Definition
A string is a finite sequence of symbols selected from an alphabet.
Represented as

$$w \in \Sigma^*$$

Example

$$\Sigma = \{0, 1\}$$

Possible strings

$$0,\ 1,\ 01,\ 101,\ 1110$$

Properties

- Order is important.
- Length is finite.
- Symbols must belong to the alphabet.

Types of Strings

1. Single Symbol String
   Example: a
2. Multiple Symbol String
   Example: abc
3. Empty String

$$\epsilon$$

   Contains zero symbols.
4. Repeated Symbol String
   Examples

$$aaaa, \ 0000$$

Valid and Invalid Strings

Given

$$\Sigma = \{0, 1\}$$

Valid Strings

$$0, \ 101, \ 1100$$

Invalid Strings

$$2, \ 102, \ abc$$

Reason:
These contain symbols that are not present in the given alphabet.

Key Idea

Every symbol in a valid string must belong to the alphabet.

Empty String (ϵ)

Definition
An empty string is a string that contains zero symbols.

Represented by

$$\epsilon$$

Characteristics

- Contains no symbols
- Length is zero

$$|\epsilon| = 0$$

- Valid over every alphabet
- Different from the empty language ($\emptyset$)

Example
Given

$$\Sigma = \{0, 1\}$$

Possible strings:

$$0,\ 1,\ 01,\ 101,\ \epsilon$$

Empty String (ϵ) vs Empty Language ($\emptyset$)

Empty String (ϵ)	**Empty Language ($\emptyset$)**
A string	A language
Contains zero symbols	Contains zero strings
Length $= 0$	Size $= 0$
One valid string	No element

Examples

$$L = \{\epsilon\}$$

(Language containing one string)

$$L = \emptyset$$

(Language containing no strings)

Important

$$\epsilon \neq \emptyset$$

Length of a String

Definition
The length of a string is the total number of symbols present in it.

Notation

$$|w|$$

Examples

$$w = abc, \qquad |w| = 3$$

$$w = 10101, \qquad |w| = 5$$

$$w = \epsilon, \qquad |\epsilon| = 0$$

Important Points

- Count every symbol.
- Order does not affect length.
- Empty string has length zero.

Examples of String Length

Given

$$\Sigma = \{a, b\}$$

Example 1

$$w = abba$$

$$|w| = 4$$

Example 2

$$w = aaaaaa$$

$$|w| = 6$$

Example 3

$$w = \epsilon$$

$$|\epsilon| = 0$$

General Rule

If

Prefix, Suffix and Substring

Let

$$w = abcd$$

Prefix

$$\epsilon,\ a,\ ab,\ abc,\ abcd$$

Suffix

$$\epsilon,\ d,\ cd,\ bcd,\ abcd$$

Substring

$$a,\ ab,\ bc,\ abc,\ bcd,\ cd$$

Key Idea

Every prefix, suffix and substring is derived from the original string.

String Concatenation

Definition
Concatenation joins two strings to form a new string.

Example

$$x = abc$$

$$y = def$$

Then

$$xy = abcdef$$

Property

$$|xy| = |x| + |y|$$

Result

abc + def → abcdef

Properties of String Concatenation

Let x, y and z be strings.

1. Closure

$$x, y \in \Sigma^* \Rightarrow xy \in \Sigma^*$$

2. Associative

$$(xy)z = x(yz)$$

3. Identity

$$x\epsilon = \epsilon x = x$$

4. Non-Commutative

$$xy \neq yx$$

Example

$$ab \neq ba$$

Definition
The reverse of a string is obtained by writing its symbols in the opposite order.
Notation

$$w^R$$

Examples

$$w = abc$$

$$w^R = cba$$

$$w = 10110$$

$$w^R = 01101$$

Special Case

$$\epsilon^R = \epsilon$$

String Equality

Two strings are equal if

- Same symbols
- Same order
- Same length

Examples

$$abc = abc$$

$$abc \neq acb$$

(Order differs)

$$abc \neq abcd$$

(Length differs)

Applications of Strings in Compiler Design

Applications

1. Lexical Analysis
2. Identifier Recognition
3. Keyword Recognition
4. Pattern Matching using Regular Expressions

Example

int number = 10;

Tokens

$\boxed{\texttt{int}}$ $\boxed{\texttt{number}}$ $\boxed{\texttt{10}}$

Key Idea

Compilers process programs as sequences of symbols and strings.

Applications of Strings

Strings are used in almost every computer application.

1. Password Systems
   - P@ss123
   - Abc@123
2. Search Engines
   - Theory of Computation
   - Artificial Intelligence
3. DNA Sequencing
   - DNA String: AGCTTAGC
   - Alphabet: $\Sigma = \{A, C, G, T\}$
4. Programming Languages
   - `printf("Hello");`
5. Text Processing
   - Spell Checking
   - Auto Correction
   - Text Searching
   - Natural Language Processing (NLP)

Key Idea

Strings form the basis of modern computing applications.

Activity: Identify Alphabet and Valid Strings
Given

$$\Sigma = \{a, b\}$$

Valid Strings

$$ab, \quad aabb, \quad bba$$

Invalid Strings

$$abc, \quad aba1, \quad hello$$

Discussion

- Which symbols belong to Σ?
- Why are some strings invalid?
- Find the length of each valid string.

Think–Pair–Share Activity

Given

$$\Sigma = \{0, 1\}$$

$$w = 101101$$

Answer the following:

1. Is w valid?
2. Find $|w|$.
3. Find w^R.
4. Concatenate w with 0.

Expected Answers

$$|w| = 6$$

$$w^R = 101101$$

$$w0 = 1011010$$

1. What is an alphabet?
2. What does ϵ represent?
3. What is the length of ϵ?
4. Is $\emptyset$ a string?
5. Find the reverse of "abc".
6. Is concatenation commutative?

Answers

- Alphabet: Finite set of symbols
- ϵ: Empty string
- $|\epsilon| = 0$
- No
- cba
- No ($ab \neq ba$)

Lecture Summary & Homework

Today's Learning

- Alphabet (Σ)
- Symbols
- Strings
- Valid and Invalid Strings
- Empty String (ϵ)
- Empty Language ($\emptyset$)
- Length of String
- Prefix, Suffix and Substring
- String Operations

Homework

1. For $\Sigma = \{a, b\}$, write all strings of length 2.
2. For $w = \texttt{computer}$, find:
    - Length
    - Reverse
    - Prefixes
    - Suffixes
3. Differentiate ϵ and $\emptyset$.

Next Lecture

Operations on Strings and Introduction to Formal Languages.